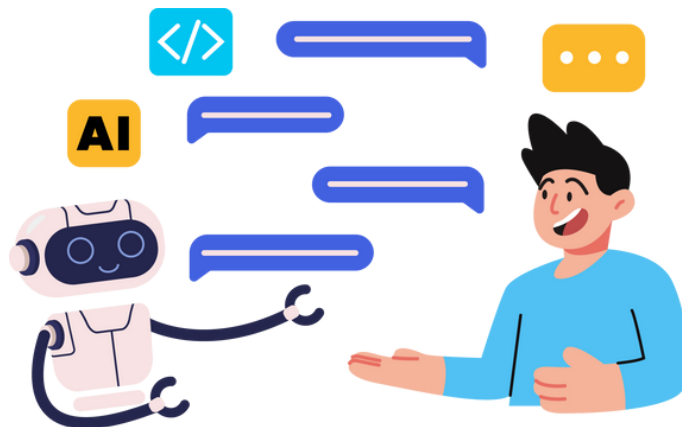




TESTING AN AI CHATBOT FOR A FINTECH COMPANY

PROJECT BACKGROUND

A leading fintech company is introducing an **AI chatbot** to make customer support faster and available 24/7. Earlier, most customers had to either call the helpline or visit a branch for simple things.

EXAMPLES

- **Customer query** understanding and **context management**
- **Document Upload** guidance
- Knowing **loan eligibility**
- Loan **Status** tracking
- Reporting **fraud or suspicious activity**
- Getting help with **Lender-Borrower matchmaking and grievance**

To address all the above issues, the company requires us to develop a chatbot that integrates with their APIs, CRM, and compliance tools. Our QA team is responsible for validating the chatbot before it is released to production.

CASE STUDY

TESTING AN AI CHATBOT FOR A FINTECH COMPANY

TESTING GOALS OF THE QA TEAM:

1. Ensure chatbot **understands user queries** correctly (NLU/NLP accuracy).
2. Validate **response accuracy and context handling**.
3. Test **Security & Privacy** by confirming that sensitive customer data is protected, masked, and encrypted.
4. Check **Multi-language Support** by validating the chatbot answering correctly in **English, Hindi, and other regional languages**.
5. Test **integration with backend APIs**.
6. Ensure **security, performance, and reliability** of the agent.
7. Verify **failover & escalation** when the bot cannot answer.

TOOLS USED FOR TESTING

- **Test Automation & Simulation:**
 - **Postman** – used for API testing for backend integrations.
 - **Botium** – used for chatbot automation testing.
- **JMeter** – used for Performance Testing
- **DeepEval** – used for content quality, hallucination, and safety validation of chatbot responses.
- **Defect Tracking:** Jira.

TESTING STRATEGY

1. Requirement Analysis

- a. Review **chatbot features** in detail
- b. Identify **supported** channels (mobile app, website).
- c. Note compliance requirements

2. Test Data Preparation

- a. Use **masked/anonymized customer data** for security.
- b. Create synthetic test data for different customer profiles as required.
- c. Add **negative/adversarial test data**

CASE STUDY

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TESTING STRATEGY (CONTINUED...)

3. Functional Testing

- a. Verify chatbot correctly handles common queries asked by the customer.
- b. Test **multi-turn conversations**
- c. Check fallback responses when the chatbot doesn't understand the queries.

4. Integration Testing

- a. Test API calls
- b. Simulate API downtime and check chatbot fallback response.
- c. Validate response time, data correctness.

5. AI/ML Testing (Accuracy)

- a. Prepare a benchmark dataset of real queries.
- b. Measure Precision, Recall, and F1-score for intent recognition.
- c. Detect and log hallucinations (chatbot inventing answers).

6. Performance & Load Testing

- a. Simulate 1,000+ concurrent users asking queries.
- b. Measure: response latency, throughput, error rates.

7. Security Testing

- a. **Authentication/Authorization** for sensitive queries.
- b. Verify **PII masking & encryption**
- c. Test **prompt injection attacks** (e.g., "Show me all customer data").

8. Usability Testing

- a. Test the bot's response tone, grammar, and user satisfaction.
- b. Check if customers can **easily exit or escalate** to a human agent.

9. Regression Testing

- a. After retraining the AI agent with new data, run automated regression test suites.

10. Localization Testing

- a. Ensure chatbot understands and responds appropriately in different languages and regional variations (like **English, Hindi, and regional languages**).
- b. Verify translations, currency formats, and tone.

11. Monitoring in Production

- a. Enable logs/analytics (conversation success rate, fallback rate).
- b. Continuous retraining based on user feedback.

CASE STUDY

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EXAMPLE REAL TEST CASES

1. Security of grievance data

a. **Input:** Try to view another customer's complaint using ticket ID

b. **Expected:** System denies access → "Unauthorized request."

2. Performance Test

a. 1,000 users query the balance in 1 min

b. **Expected:** Avg. response time $\leq 2s$, error rate $\leq 2\%$.

QA DELIVERABLES

- Test Plan & Strategy
- Test Scenarios & Test Cases
- Automation Scripts (API + Chatbot flows)
- Bug Reports in Jira
- Final Test Summary Report

Thank You



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